$$r(t) = \langle \mathcal{E}_-(t)|\mathcal{E}_+(t)\rangle = \prod_i \exp\left[-\frac{1}{2}\left|\lambda_i(t)\right|^2\right]$$
$$= \exp\left[-\sum_i 4\frac{\left|g_i\right|^2}{\omega_i^2}\left(1-\cos\omega_i t\right)\right]. \tag{5.108}$$

For times t much smaller than the dynamical timescales ω_i^{-1} of the environmental modes, i.e., $\omega_i t \ll 1$, we can approximate the cosine in (5.108) by

$$\cos\omega_i t \approx 1 - \frac{1}{2}\omega_i^2 t^2. \tag{5.109}$$